



Prusa Core One Drybox

I Ivars

[VIEW IN BROWSER](#)

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Summary

A drybox created by a polycarbonate window in a magnetic frame and a magnetic desiccant box.

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Tags: [box](#) [dry](#) [one](#)

Materials Needed

11 20x6x2 mm N52 magnets (don't get weaker magnets) <https://www.aliexpress.com/i/3256806665685488.html?gQT=1&gatewayAdapt=4itemAdapt>
double sided mounting tape https://www.amazon.com/dp/B08ZXS4PS1?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1
sealing tape https://www.amazon.com/dp/B07L6KV4QG?ref=ppx_yo2ov_dt_b_fed_asin_title
1/16x12x12 inch smoked polycarbonate sheet https://www.amazon.com/dp/B09SFC8J56?ref=ppx_yo2ov_dt_b_fed_asin_title
medium gap filling CA https://www.amazon.com/Pacer-Technology-2oz-Zap-Gap/dp/B079VNCXV5/ref=pd_sbs_d_sccl_2_4/137-0777470-0675619?pd_rd_w=jnG3K&content-id=amzn1.sym.7cf15969-68a2-473c-b74e-2afd518e17cd&pf_rd_p=7cf15969-68a2-473c-b74e-2afd518e17cd&pf_rd_r=F8BCTXY51Z6N7KMWMAQS&pd_rd_wg=SPd7k&pd_rd_r=a623-b6bcb4356b8&pd_rd_i=B079VNCXV5&psc=1

CA accelerator (kicker) [https://www.amazon.com/Bob-Smith-Industries-BSI-151H-Accelerator/dp/B0166FFCCS/ref=sr_1_2?](https://www.amazon.com/Bob-Smith-Industries-BSI-151H-Accelerator/dp/B0166FFCCS/ref=sr_1_2?crid=6973HX21UC64&dib=eyJ2ljojMSJ9.KI9wq3Xz78drDi9XEu8FdMDoZhRd1oQIvw7P3I)

[crid=6973HX21UC64&dib=eyJ2ljojMSJ9.KI9wq3Xz78drDi9XEu8FdMDoZhRd1oQIvw7P3I](https://www.amazon.com/Bob-Smith-Industries-BSI-151H-Accelerator/dp/B0166FFCCS/ref=sr_1_2?crid=6973HX21UC64&dib=eyJ2ljojMSJ9.KI9wq3Xz78drDi9XEu8FdMDoZhRd1oQIvw7P3I)
CsmolFtMbT3ninJx4j5Wh6ACU7aqYV0AKdLO7werpN_IER658WTKmgroXxVECVG8FfUabC
humidity gauge

humidity gauge [https://www.amazon.com/dp/B08PVLf8LD?](https://www.amazon.com/dp/B08PVLf8LD?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_1)

[ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_1](https://www.amazon.com/dp/B08PVLf8LD?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_1)

dryer vent aluminum foil tape [https://www.amazon.com/Aluminum-](https://www.amazon.com/Aluminum-Ductwork-Temperature-Insulation-Adhesive/dp/B0B6NDFXPN/ref=sr_1_4?crid=2FZ0Q95T2DXWP&dib=eyJ2ljojMSJ9._JENdIS0VzOjmcnHduiL3kOMKhzriBbDGmP9h)

[Ductwork-Temperature-Insulation-Adhesive/dp/B0B6NDFXPN/ref=sr_1_4?](https://www.amazon.com/Aluminum-Ductwork-Temperature-Insulation-Adhesive/dp/B0B6NDFXPN/ref=sr_1_4?crid=2FZ0Q95T2DXWP&dib=eyJ2ljojMSJ9._JENdIS0VzOjmcnHduiL3kOMKhzriBbDGmP9h)

[crid=2FZ0Q95T2DXWP&dib=eyJ2ljojMSJ9._JENdIS0VzOjmcnHduiL3kOMKhzriBbDGmP9h](https://www.amazon.com/Aluminum-Ductwork-Temperature-Insulation-Adhesive/dp/B0B6NDFXPN/ref=sr_1_4?crid=2FZ0Q95T2DXWP&dib=eyJ2ljojMSJ9._JENdIS0VzOjmcnHduiL3kOMKhzriBbDGmP9h)
zDcisFtbUXkaigKtTdtYMH9eDPA7zWiRThyxKXQWGcymPPaY5cKA6QMsz6FFoSIEF-8R4Q4

[GjGTm0RBE2ar-lwlJrS8bCl_zbsoALIYEA_d1spZ8a7kdetlhwxxc-](https://www.amazon.com/Aluminum-Ductwork-Temperature-Insulation-Adhesive/dp/B0B6NDFXPN/ref=sr_1_4?crid=2FZ0Q95T2DXWP&dib=eyJ2ljojMSJ9._JENdIS0VzOjmcnHduiL3kOMKhzriBbDGmP9h)

[dPdIpbveUMqM8xf_PUaPG9UHQbE9QXzejU2E51Kzns3Q.TS1vPf1zRen-8IYwStHnX3Dd9T](https://www.amazon.com/Aluminum-Ductwork-Temperature-Insulation-Adhesive/dp/B0B6NDFXPN/ref=sr_1_4?crid=2FZ0Q95T2DXWP&dib=eyJ2ljojMSJ9._JENdIS0VzOjmcnHduiL3kOMKhzriBbDGmP9h)

Dimensions

I bought 2 polycarbonate sheets. The first one fit perfectly and I used it to practice cutting an opening for the humidistat. All did not go well and I had to use the second sheet. To my surprise it was significantly undersize so I had to make some dimensional adjustments and reprint. With that in mind, I have included .step files and .f3d Fusion files.

Construction Notes

- The frame will have to be printed with supports. Two corners will fit on the build plate. I used Prusament galaxy black PLA.

- The handle should be printed with the opening toward the build plate so supports are not needed.

- I found it best to glue the sides together and then join the sides to form the frame.

- To glue the sides, clamp a straight board to a flat surface. Carefully slide the two corners together (face up) along the board. It is helpful to have an other person spray the kicker while you hold the pieces firmly against the board and flat surface.

- Glue in the magnets and apply the sealing tape. The tape can be cut accurately by placing a sharp knife at the seam and tearing off the tape against the blade.

- Cut the hole for the humidity gauge using the templates. I used a jig saw. The gauge will fit tightly and some sanding is necessary. Apply the double sided tape and press the polycarbonate into the frame firmly and evenly.

- Mount the gauge in the polycarbonate. In retrospect I would simply glue the gauge to the polycarbonate.

- Glue the handle and you are done with the frame.

- Print the desiccant box (you will have to flip the lid in the slicer). The lid fits tightly and it is easier to snap it into place by first sliding the corner

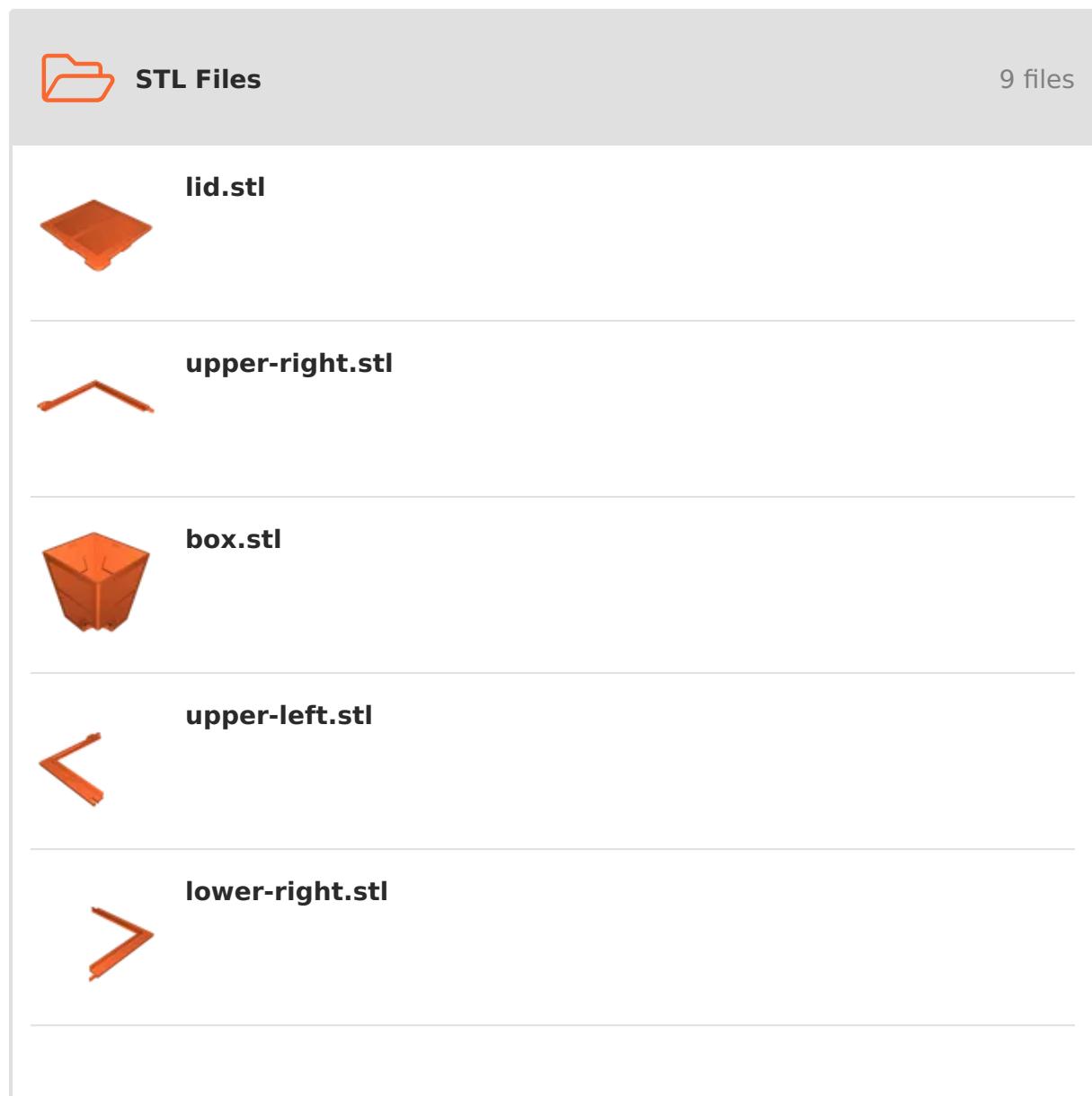
with the tab into the corresponding corner of the box.

-Use the dryer vent tape to seal the spool indent. It is easier (and looks better) to tape the corner seams from the inside of printer. See last picture. Okay I lied - the lower seam in the back is hard! Shine a flashlight on the inside and see if there is any light showing on the outside. There is a hole on the right side of the spool indent that should also be sealed.

-Enjoy your handiwork!

Regrets-I wish the polycarbonate was a little darker.-I think the desiccant box would look better in galaxy black. Maybe I'll do that after I finish my upcoming projects.

Model files





lower-left.stl



handle.stl



humidistat-cut-template.stl



humidistat-drill-template.stl



STEP Files

2 files



drybox-frame.step



drybox-dessicant-box.step

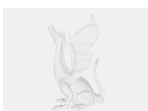


Fusion Files

2 files



drybox-frame.f3d



drybox-dessicant-box.f3d

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